Simplified GDPR Consent Manager

Branch Coverage Report

Decision (branch) coverage - every if/else, ?:, &&, ||, ?? and default-value path

#	Name	SRN
1	SUMAN DIDDIMANI	PES1UG24AM409
2	PRADEEP T	PES1UG24AM385
3	VILAS	PES1UG24AM415
4	PRANAAV PRABHAKARAN	PES1UG24AM387

Version	Date	Author	Change Summary
1.0	03-11-2026	Team 6	Report generated from the executed test run (Sprint 3 testing, issue #25)

1. Summary

Coverage was measured on 03-11-2026 while running the full unit test suites (server: Jest/Istanbul, client: Vitest/V8). Target from the Test Plan (NFR-12, TC-MAINT-01): at least 70 % on the service layer.

Part	Branches covered	Total branches	Branch coverage
Server API	262	296	88.51 %
Client UI	332	389	85.34 %
Combined	594	685	86.72 %

2. Coverage by architectural layer

Part	Layer	Covered / total branches	Coverage
Server	Other	8/9	88.9 %
Server	Shared libraries	19/23	82.6 %
Server	Middleware (security, auth, errors)	38/41	92.7 %
Server	Service / DAO	143/163	87.7 %
Server	API routes (controllers)	9/10	90.0 %
Server	Validation (Zod schemas)	13/14	92.9 %
Server	Scripts (seed, backup, restore)	32/36	88.9 %
Client	Other	1/4	25.0 %
Client	API client	12/18	66.7 %
Client	Auth context	25/26	96.2 %
Client	Components	81/101	80.2 %
Client	Pages	213/240	88.8 %

Service layer branches coverage: 87.7 % - target 70 % met.

3. Server files

File	Covered	Total	Branch %
app.ts	2	2	100.0
config.ts	2	3	66.7
lib/asyncHandler.ts	0	0	100 (none)
lib/emailCrypto.ts	2	2	100.0
lib/logger.ts	5	6	83.3
lib/prisma.ts	0	0	100 (none)
lib/retention.ts	2	4	50.0
lib/session.ts	4	5	80.0
lib/staticClient.ts	6	6	100.0
middleware/errorHandler.ts	4	4	100.0
middleware/requireAuth.ts	5	5	100.0
middleware/requireRole.ts	7	9	77.8
middleware/security.ts	22	23	95.7
modules/audit/audit.service.ts	13	13	100.0
modules/audit/integrity.service.ts	8	9	88.9
modules/auth/auth.routes.ts	1	1	100.0
modules/auth/auth.schemas.ts	0	0	100 (none)
modules/auth/auth.service.ts	29	33	87.9
modules/consents/consents.routes.ts	0	0	100 (none)
modules/consents/consents.schemas.ts	0	0	100 (none)
modules/consents/consents.service.ts	21	22	95.5
modules/notices/notices.routes.ts	0	0	100 (none)
modules/notices/notices.service.ts	12	15	80.0
modules/policies/policies.routes.ts	3	3	100.0
modules/policies/policies.schemas.ts	0	0	100 (none)
modules/policies/policies.service.ts	11	17	64.7
modules/purposes/purposes.routes.ts	2	2	100.0
modules/purposes/purposes.schemas.ts	2	2	100.0
modules/purposes/purposes.service.ts	6	7	85.7
modules/query/query.routes.ts	3	4	75.0
modules/query/query.schemas.ts	6	7	85.7
modules/query/query.service.ts	23	26	88.5
modules/users/admin.service.ts	10	11	90.9
modules/users/timeTravel.service.ts	10	10	100.0
modules/users/users.routes.ts	0	0	100 (none)
modules/users/users.schemas.ts	5	5	100.0
routes/health.ts	4	4	100.0
scripts/backup.ts	3	4	75.0
scripts/encryptEmails.ts	0	1	0.0
scripts/restore.ts	23	23	100.0
scripts/seed.ts	6	8	75.0
types/express.d.ts	0	0	100 (none)

4. Client files

File	Covered	Total	Branch %
/home/claude/skeleton/client/postcss.config.js	0	1	0.0
App.tsx	1	1	100.0
api/auth.ts	9	11	81.8
api/client.ts	0	1	0.0
api/consents.ts	1	2	50.0
api/dpo.ts	1	2	50.0
api/notices.ts	1	2	50.0
api/policies.ts	0	0	100 (none)
api/users.ts	0	0	100 (none)
auth/AuthContext.tsx	11	12	91.7
auth/RequireAuth.tsx	10	10	100.0
auth/roles.ts	4	4	100.0
components/AdminNav.tsx	4	4	100.0
components/AuthCard.tsx	2	2	100.0
components/FormField.tsx	5	9	55.5
components/IntegrityCheck.tsx	13	16	81.2
components/Layout.tsx	11	11	100.0
components/Pager.tsx	4	4	100.0
components/PolicyNoticeBanner.tsx	18	26	69.2
components/PolicyStatusBadge.tsx	1	1	100.0
components/StatusPill.tsx	7	9	77.8
components/TimeTravel.tsx	16	19	84.2
main.tsx	0	1	0.0
pages/AdminPoliciesPage.tsx	36	42	85.7
pages/AdminUsersPage.tsx	24	32	75.0
pages/DashboardPage.tsx	44	45	97.8
pages/DpoConsolePage.tsx	48	52	92.3
pages/HistoryPage.tsx	19	20	95.0
pages/HomePage.tsx	2	3	66.7
pages/LoginPage.tsx	17	19	89.5
pages/NotFoundPage.tsx	1	1	100.0
pages/PolicyPage.tsx	7	10	70.0
pages/RegisterPage.tsx	15	16	93.8
/home/claude/skeleton/client/tailwind.config.js	0	1	0.0

5. Branches not covered (server)

Each row is one decision outcome that no test executed. Most are defensive fall-backs (e.g. 'value ?? default', missing headers, production-only paths).

File	Line	Type	Missed path
config.ts	38	if	path 1 of 1
lib/logger.ts	26	cond-expr	path 1 of 2
lib/retention.ts	9	default-arg	path 1 of 1
lib/retention.ts	16	default-arg	path 1 of 1
lib/session.ts	27	if	path 1 of 1
middleware/requireRole.ts	27	binary-expr	path 2 of 2
middleware/requireRole.ts	52	if	path 1 of 1
middleware/security.ts	19	binary-expr	path 2 of 2
modules/audit/integrity.service.ts	69	cond-expr	path 2 of 2
modules/auth/auth.service.ts	45	if	path 1 of 1
modules/auth/auth.service.ts	45	binary-expr	path 1 of 2
modules/auth/auth.service.ts	45	binary-expr	path 2 of 2
modules/auth/auth.service.ts	135	cond-expr	path 1 of 2
modules/consents/consents.service.ts	167	default-arg	path 1 of 1
modules/notices/notices.service.ts	34	if	path 1 of 1
modules/notices/notices.service.ts	53	if	path 1 of 1
modules/notices/notices.service.ts	98	if	path 1 of 1
modules/policies/policies.service.ts	23	if	path 1 of 1
modules/policies/policies.service.ts	113	binary-expr	path 2 of 2
modules/policies/policies.service.ts	116	cond-expr	path 1 of 2
modules/policies/policies.service.ts	116	cond-expr	path 2 of 2
modules/policies/policies.service.ts	117	if	path 1 of 1
modules/policies/policies.service.ts	120	if	path 1 of 1
modules/purposes/purposes.service.ts	8	default-arg	path 1 of 1
modules/query/query.routes.ts	57	cond-expr	path 2 of 2
modules/query/query.schemas.ts	41	if	path 1 of 1
modules/query/query.service.ts	45	default-arg	path 1 of 1
modules/query/query.service.ts	78	default-arg	path 1 of 1
modules/query/query.service.ts	136	binary-expr	path 2 of 2
modules/users/admin.service.ts	87	if	path 1 of 1
scripts/backup.ts	71	default-arg	path 1 of 1
scripts/encryptEmails.ts	15	if	path 1 of 1
scripts/seed.ts	75	default-arg	path 1 of 1
scripts/seed.ts	102	binary-expr	path 2 of 2

6. Branches not covered (client)

File	Line	Type	Missed path
/home/claude/skeleton/client/postcss.config.js	1	branch	path 1 of 1
api/auth.ts	50	branch	path 1 of 1
api/auth.ts	50	branch	path 1 of 1
api/client.ts	1	branch	path 1 of 1

File	Line	Type	Missed path
api/consents.ts	19	branch	path 1 of 1
api/dpo.ts	37	branch	path 1 of 1
api/notices.ts	12	branch	path 1 of 1
auth/AuthContext.tsx	59	branch	path 1 of 1
components/FormField.tsx	8	branch	path 1 of 1
components/FormField.tsx	16	branch	path 1 of 1
components/FormField.tsx	19	branch	path 1 of 1
components/FormField.tsx	25	branch	path 1 of 1
components/IntegrityCheck.tsx	7	branch	path 1 of 1
components/IntegrityCheck.tsx	38	branch	path 1 of 1
components/IntegrityCheck.tsx	25	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	48	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	62	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	63	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	66	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	94	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	24	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	31	branch	path 1 of 1
components/PolicyNoticeBanner.tsx	76	branch	path 1 of 1
components/StatusPill.tsx	27	branch	path 1 of 1
components/StatusPill.tsx	30	branch	path 1 of 1
components/TimeTravel.tsx	93	branch	path 1 of 1
components/TimeTravel.tsx	118	branch	path 1 of 1
components/TimeTravel.tsx	34	branch	path 1 of 1
main.tsx	1	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	132	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	40	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	53	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	65	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	70	branch	path 1 of 1
pages/AdminPoliciesPage.tsx	91	branch	path 1 of 1
pages/AdminUsersPage.tsx	88	branch	path 1 of 1
pages/AdminUsersPage.tsx	165	branch	path 1 of 1
pages/AdminUsersPage.tsx	30	branch	path 1 of 1
pages/AdminUsersPage.tsx	47	branch	path 1 of 1
pages/AdminUsersPage.tsx	137	branch	path 1 of 1
pages/AdminUsersPage.tsx	140	branch	path 1 of 1
pages/AdminUsersPage.tsx	156	branch	path 1 of 1
pages/AdminUsersPage.tsx	152	branch	path 1 of 1
pages/DashboardPage.tsx	105	branch	path 1 of 1
pages/DpoConsolePage.tsx	183	branch	path 1 of 1
pages/DpoConsolePage.tsx	262	branch	path 1 of 1
pages/DpoConsolePage.tsx	239	branch	path 1 of 1
pages/DpoConsolePage.tsx	251	branch	path 1 of 1
pages/HistoryPage.tsx	45	branch	path 1 of 1
pages/HomePage.tsx	43	branch	path 1 of 1
pages/LoginPage.tsx	33	branch	path 1 of 1
pages/LoginPage.tsx	33	branch	path 1 of 1
pages/PolicyPage.tsx	32	branch	path 1 of 1

File	Line	Type	Missed path
pages/PolicyPage.tsx	51	branch	path 1 of 1
pages/PolicyPage.tsx	20	branch	path 1 of 1
pages/RegisterPage.tsx	40	branch	path 1 of 1
/home/claude/skeleton/client/tailwind.config.js	1	branch	path 1 of 1

How to reproduce

- npm run test:coverage -w server -> server/coverage/lcov-report/index.html
- npx vitest run --coverage (client, @vitest/coverage-v8) -> client/coverage/index.html
- The HTML reports (per-line highlighting) are included in docs/07-Test-Reports/raw/.